A Computable Spectral-Layer Occupancy Closure on a Projected Two-Center Background

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Abstract

We present a compact, journal-facing version of the Projection Spectral Layer Theory (PSLT) as a conditional EFT-level spectral-layer occupancy closure on a fixed projected two-center conformal background. The central object is a normalized layer probability $P_N = W_N/\sum_K W_K$, where W_N combines an effective phase-space weight g_N, a rank-2 kinetic rate Γ_N, and a visibility factor B_N. The claim of this short article is deliberately calibrated: under the stated assumptions and finite audit domains, the release baseline shows first-three-layer occupancy concentration and a finite-domain no-fourth-bound-layer kinetic certificate. It is not presented as a complete EYMH derivation of exactly three Standard Model fermion generations, nor as an absolute Higgs-sector prediction. The long technical manuscript is treated as a supplementary audit companion containing the proposition chains, certificates, artifact registry, and negative/diagnostic branches.

1 Introduction and Claim Calibration

The Standard Model accommodates three fermion families but does not determine the family count dynamically. Traditional flavor, GUT, and compactification approaches organize charges or correlations among Yukawa couplings, while the family multiplicity is often supplied as an external input or model-building choice [1–12]. PSLT asks a narrower question: whether a projected geometric spectral problem, combined with a specified EFT closure, yields a reproducible concentration of observable weight in the first three spectral layers.

The reason for this narrower formulation is not merely rhetorical. In a conventional family-count theorem one would first identify a microscopic Hilbert space of chiral Standard Model fermions and then prove that its protected low-energy multiplicity is exactly three. The current PSLT release does not yet provide that theorem. Instead, it supplies a computable chain

$$\Omega(\rho, z; D_{\mathrm{sep}}) \longrightarrow V_{\mathrm{eff}} \longrightarrow \{\omega_N, S_N\} \longrightarrow \Gamma_N \longrightarrow P_N, \tag{1}$$

and asks whether this chain is internally stable, reproducible, and physically interpretable under explicitly stated assumptions. The distinction matters for review: the evidence presented below supports the conditional closure in Eq. (1); it should not be read as a hidden proof of an ultraviolet completion.

1.1 Relation to familiar flavor and unification programs

Gauge-unification frameworks such as $SU(5)$, $SO(10)$, Pati–Salam, E_6, and related group-theoretic embeddings organize one Standard Model family into larger representations and constrain charge assignments. Flavor-symmetry models organize observed mixing patterns and Yukawa hierarchies. String or geometric compactifications can produce family multiplicities through topological data, but the multiplicity is then tied to the chosen compactification and bundle sector. The present work is deliberately orthogonal to those goals. It does not propose a new charge-unification group and does not replace CKM/PMNS flavor phenomenology. It proposes a projected spectral-layer occupancy model whose output is a normalized scan weight.

This positioning is important because the same integer "three" can mean three different things:

1. three unpaired chiral Standard Model families in a microscopic fermion theory;
2. three bound or low-lying layers of a projected scalar spectral operator;
3. first-three-layer dominance of a normalized EFT occupancy measure.

Only the third item is the baseline claim of this short article. The first item is reserved for a future protected-index completion, and the second item is not generally true at every audited value of D_{sep}: the finite-domain kinetic proxy certifies absence of a fourth bound layer, not the existence of exactly three bound modes everywhere.

This short article is designed to be read independently of the long technical companion. Its physics claims are:

1. A fixed projected two-center conformal background defines a computable scalar spectral operator and WKB barrier data.
2. A rank-2 two-center kinetic closure maps localized spectral information into $\Gamma_N(D_{\mathrm{sep}}, \eta)$.
3. A bounded effective phase-space weight g_N, kinetic rate Γ_N, and visibility factor B_N define the scan-level occupancy probability P_N.
4. On the audited release grid, the first-three-layer occupancy ratio $\mathcal{R}_3 = P_1 + P_2 + P_3$ is concentrated over most of the sampled domain, and the audited finite-volume kinetic proxy has no fourth bound layer.
5. The $H \to \mu\mu$ map is a reference-normalized diagnostic target-region comparison to ATLAS data [13], not a formal global likelihood, exclusion, or absolute Higgs-sector prediction.

The manuscript does *not* claim a full action-level theorem proving exactly three Standard Model generations. Optional protected Spin^c, DSS/Floquet, open-system, and static-width packages are treated as supplementary or diagnostic material unless explicitly promoted in a future revision.

1.2 What would falsify or weaken the present closure

Because the present article is a conditional computational closure, the relevant failure modes are also concrete. The release would be weakened if the live figure-generation path failed to reproduce the registered $\mathcal{R}_3$ fractions, if the WKB table failed the relation $r_1 = \exp(-2S_1)$, if the high-N kinetic audit supported a fourth bound layer on the certified domain, or if the reference-normalized Higgs diagnostic were described as an absolute prediction. Conversely, none of the following would by itself invalidate the present short claim: failure of the optional DSS/Floquet parent attempt, non-promotion of global static-width poles, or the absence of a complete protected Spin^c family-count theorem. Those are future strengthening routes, not assumptions silently used in the release numbers.

2 Assumption Map

All propositions in this short article are conditional on the assumptions in Table 1. The purpose of the table is to prevent the short article from overclaiming what is actually established by the current audited chain. The assumptions are not hidden fit knobs. They are the boundary conditions under which the release figures, tables, and certificates are to be interpreted. In particular, the finite-domain numerical checks certify the closure *inside* this assumption map; they do not erase the need for a future parent derivation or a future microscopic flavor theorem.

Table 1: Submission assumption map for the short-article manuscript.

Label	Assumption	Operational reading
A1	Fixed projected two-source conformal background.	The conformal factor is the projected Poisson/Plummer two-center solution; deriving the same source from a complete higher-dimensional EYMH parent is future work.
A2	Spectral label as EFT generation-layer label.	The scalar spectral index N labels scan layers; it is not by itself a proof of SM chiral family multiplicity.
A3	Effective phase-space weight.	The release g_N is a bounded two-dimensional phase-space/shell-volume profile with low-mode certificates and a high-N tail prescription; Cardy-style formulas are motivational comparators only.
A4	Rank-2 kinetic closure.	The two-center localized subspace gives a coarse-grained 2×2 growth matrix; strict static-width promotion remains local to its certified parent tube.
A5	Reference-normalized visibility and Higgs diagnostic.	B_N and $H \to \mu\mu$ define scan-level EFT/Wilson-inspired diagnostics, not absolute experimental validation.

2.1 Dependency of the claims on A1–A5

The role of each assumption can be summarized as follows. A1 fixes the operator domain and the geometry entering V_{eff}. A2 is the interpretive bridge allowing the scalar index N to be used as an EFT layer label in the scan. A3 supplies the effective shell weight multiplying each layer. A4 supplies the kinetic time scale through Γ_N. A5 supplies the map from the occupancy variables to visible-sector diagnostics. Thus the baseline theorem schema is not

$$S_{\text{EYMH}} \Longrightarrow N_{\text{SM}} = 3, \tag{2}$$

but rather

$$\begin{aligned} &\text{A1–A5 plus audited finite domains} \\ &\quad\Longrightarrow \text{first-three-layer concentration} \\ &\qquad \text{and no audited fourth bound layer.} \end{aligned} \tag{3}$$

This schema is intentionally conservative. For example, the optional protected Spin^c route would replace part of A2 by an index-theoretic family-count theorem, while a future higher-dimensional EYMH derivation would replace A1 by a parent-side result. Neither replacement is assumed in the numbers reported here.

Table 2: Where each assumption enters the short-article chain.

Assumption	First equation affected	Failure mode if removed
A1	Ω and V_{eff} in Eqs. (5)–(11)	No unique projected operator or WKB barrier.
A2	Layer label N in Eq. (10)	Occupancy layers lose generation-layer interpretation.
A3	g_N in Eq. (17)	High-N weights are undefined.
A4	Γ_N in Eqs. (19)–(22)	No kinetic formation clock for P_N.
A5	B_N and Eq. (31)	No visible-sector diagnostic map.

2.2 Parameter-set taxonomy

Several numerical parameter sets appear in the long audit companion. They should not be read as interchangeable baselines. The short article uses the release scan only for the reported occupancy and Higgs-diagnostic numbers; other parameter sets are demonstrators, validation tests, or historical comparators. Table 3 fixes this convention.

Table 3: Parameter-set taxonomy used when reading the short article together with the long supplement.

Name	**Role**	**How it may be used**
Release baseline scan	The canonical source of the reported $\mathcal{R}_3$, no-fourth, and $H \to \mu\mu$ diagnostic numbers.	Use for all headline release statements and figures in this short article.
Demonstrator geometry	Transparent worked examples of the Plummer two-center operator and WKB action convention.	Use to inspect formulas and signs; do not substitute into release fractions.
Physical-gap and finite-volume validation sets	Grid, Sturm/min–max, low-mode, and projector-coherence audits.	Use as certificates for finite-domain statements, not as separate phenomenological baselines.
Legacy comparators and stress tests	Cardy-style weights, Yukawa visibility laws, CAP/ECS/static-width trials, DSS/Floquet trials, and other diagnostic branches.	Use only for uncertainty, failure-mode, or motivation discussion unless explicitly promoted.

3 Projected Two-Center Spectral Operator

The geometric input is a static conformally flat background,

$$g_{\mu\nu}(x) = \Omega^2(x)\eta_{\mu\nu}, \tag{4}$$

with two projected centers separated by $D_{\rm sep}$. For a Plummer regulator ϵ, the working conformal factor is

$$\Omega(\rho, z; D_{\rm sep}) = 1 + a\left(\frac{1}{r_+} + \frac{1}{r_-}\right), \qquad r_\pm = \sqrt{\rho^2 + (z \mp D_{\rm sep}/2)^2 + \epsilon^2}. \tag{5}$$

It solves the projected Poisson equation with two smeared unit sources. Indeed, for

$$f_\epsilon(r) = (r^2 + \epsilon^2)^{-1/2}, \tag{6}$$

the three-dimensional radial Laplacian gives

$$\nabla^2 f_\epsilon = f_\epsilon'' + \frac{2}{r} f_\epsilon' = -\frac{3\epsilon^2}{(r^2 + \epsilon^2)^{5/2}}. \tag{7}$$

Thus, if

$$\rho_\epsilon(r) = \frac{3\epsilon^2}{4\pi(r^2 + \epsilon^2)^{5/2}}, \qquad \int_{\mathbb{R}^3} \rho_\epsilon(r)\, \mathrm{d}^3x = 1, \tag{8}$$

then Eq. (5) obeys

$$\nabla^2\Omega = -4\pi a \left[\rho_\epsilon(x - x_+) + \rho_\epsilon(x - x_-)\right]. \tag{9}$$

3.1 Conformal Klein–Gordon reduction

For a scalar test field with non-minimal curvature coupling [14], the conformal reduction gives

$$\left[-\nabla^2 + V_{\text{eff}}(x)\right]\psi_N = \omega_N^2\psi_N, \tag{10}$$

where

$$V_{\text{eff}} = m_0^2\Omega^2 + (1-6\xi)\Omega^{-1}\nabla^2\Omega. \tag{11}$$

For completeness, we give the derivation because it is one of the few analytic steps in the release chain that should be easy to audit. Start from

$$(\Box_g - m_0^2 - \xi R)\Phi = 0, \qquad \Phi(t,x) = \mathrm{e}^{-i\omega t}\Omega^{-1}(x)\psi(x). \tag{12}$$

In four dimensions,

$$\left(\Box_g - \frac{1}{6}R\right)\Omega^{-1}\psi = \Omega^{-3}\Box_\eta\psi. \tag{13}$$

Substitution gives

$$\Box_\eta\psi - m_0^2\Omega^2\psi + \left(\frac{1}{6} - \xi\right)\Omega^2 R\psi = 0. \tag{14}$$

With a static Ω and $g_{\mu\nu} = \Omega^2\eta_{\mu\nu}$,

$$R = -6\Omega^{-3}\nabla^2\Omega, \tag{15}$$

and Eq. (11) follows after separating the time dependence. This derivation is local and kinematic. It does not claim that Eq. (9) has already been obtained as the unique solution of a complete backreacted EYMH system.

3.2 Finite-volume operator used in the audits

The numerical release works with finite-volume discretizations of Eq. (10). Abstractly, let $\mathcal{D}_L$ be the audited cylindrical or one-dimensional projected domain and impose the boundary condition used by the relevant certificate. The finite-volume operator is

$$H_{D_{\text{sep}}}^{(L,h)} = -\Delta_h + V_{\text{eff}}^{(h)}(D_{\text{sep}}), \tag{16}$$

with h denoting the mesh scale. Self-adjointness at the matrix level is not optional: it is what allows the spectral ordering, Sturm/min–max counting, and Riesz-projector transport statements later in the paper to be meaningful. The short article therefore reports only certified statements that can be traced back to a specified finite-volume operator chain.

Figure 1 shows the physical geometry behind this operator.

4 Rank-2 Occupancy Closure

The scan observable is the normalized layer probability

$$P_N(t_{\text{coh}}; D_{\text{sep}}, \eta) = \frac{W_N(t_{\text{coh}}; D_{\text{sep}}, \eta)}{\sum_K W_K(t_{\text{coh}}; D_{\text{sep}}, \eta)}, \qquad W_N = B_N g_N\left(1 - \mathrm{e}^{-\Gamma_N(D_{\text{sep}},\eta)t_{\text{coh}}}\right). \tag{17}$$

The three factors have distinct roles: g_N is an effective phase-space weight, Γ_N is a kinetic formation rate, and B_N is a visibility weight. Figure 2 summarizes the short-article closure chain and the associated claim firewall.

The normalization in Eq. (17) is conditional on the chosen scan window and on the set of layers retained in the denominator. It is therefore a relative occupancy measure, not a conserved

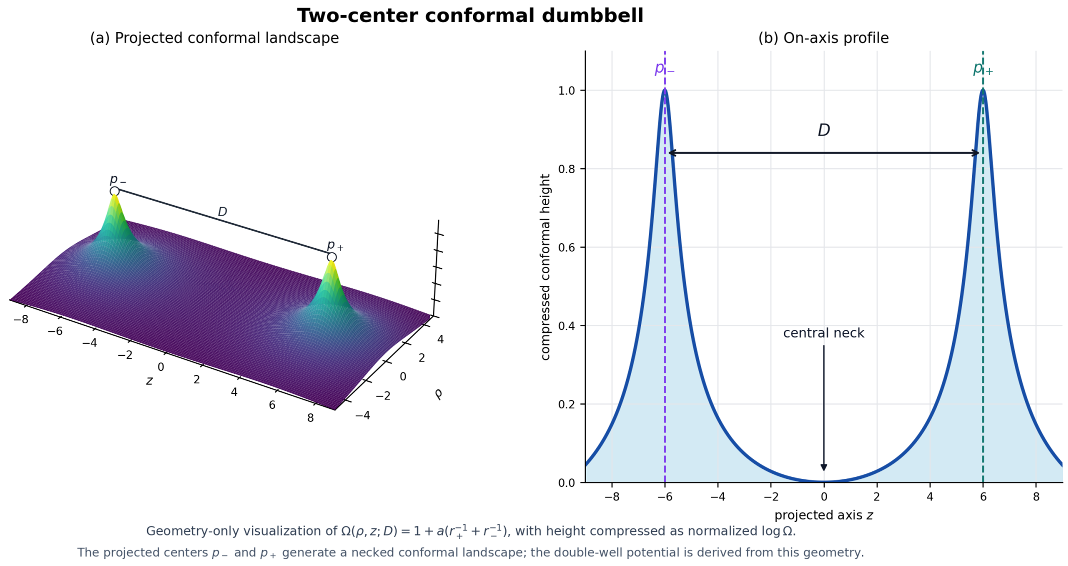

Figure 1: Projected two-center conformal dumbbell used as the fixed background in the release closure. The figure is a physical visualization of Eq. (5), not a new independent model input.

probability derived from a microscopic unitary time evolution. For small $\Gamma_N t_{\rm coh}$, the kinetic factor behaves as

$$1 - e^{-\Gamma_N t_{\rm coh}} = \Gamma_N t_{\rm coh} + \mathcal{O}((\Gamma_N t_{\rm coh})^2), \tag{18}$$

so the early-time ordering is controlled by $B_N g_N \Gamma_N$. For large $\Gamma_N t_{\rm coh}$, the same factor saturates and the ordering is controlled by $B_N g_N$. This simple interpolation is the reason $t_{\rm coh}$ is treated as a closure parameter rather than as a directly measured lifetime.

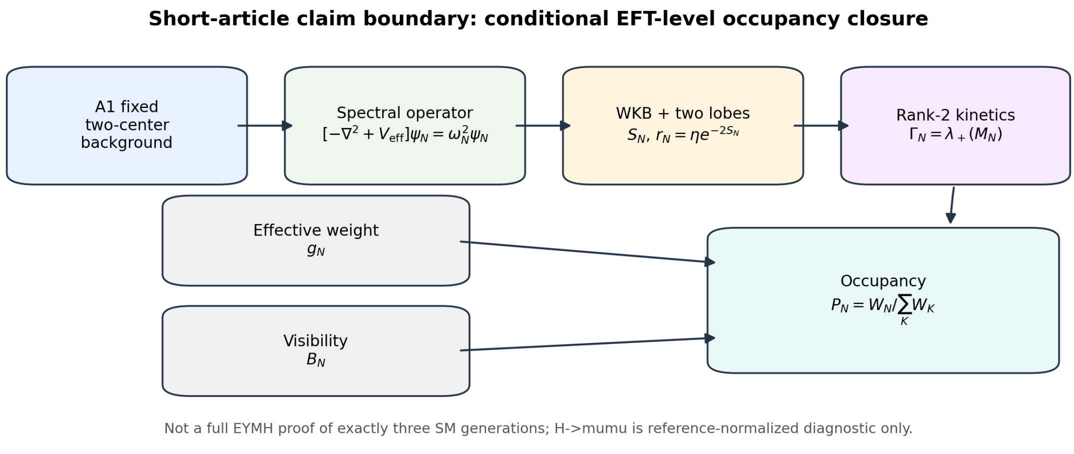

Figure 2: short-article closure diagram. The figure exposes the intended logical order: fixed projected background, spectral operator, WKB/two-lobe data, rank-2 kinetics, effective phase-space and visibility weights, and finally the normalized occupancy P_N. The bottom line is part of the claim boundary: this is a conditional occupancy closure, not a complete first-principles proof of exactly three SM generations.

For the kinetic module, the two localized lobes define an effective rank-2 subspace. A representative closure matrix is

$$M_N = r_N \begin{pmatrix} \Gamma_{N,a} & \chi\sqrt{\Gamma_{N,a}\Gamma_{N,b}} \\ \chi\sqrt{\Gamma_{N,a}\Gamma_{N,b}} & \Gamma_{N,b} \end{pmatrix}, \qquad r_N = \eta e^{-2S_N}. \tag{19}$$

The formation rate is the positive part of the largest eigenvalue,

$$\Gamma_N = \max\{0, \lambda_+(M_N)\}. \tag{20}$$

Explicitly,

$$\lambda_+(M_N) = r_N \left[\frac{\Gamma_{N,a} + \Gamma_{N,b}}{2} + \sqrt{\left(\frac{\Gamma_{N,a} - \Gamma_{N,b}}{2} \right)^2 + \chi^2 \Gamma_{N,a} \Gamma_{N,b}} \right]. \tag{21}$$

Thus M_N is positive semidefinite whenever $r_N \geq 0$, $\Gamma_{N,a,b} \geq 0$, and $|\chi| \leq 1$. When $\eta > 1$, the factor $r_N = \eta \mathrm{e}^{-2S_N}$ should not be called a literal single-particle tunneling probability; it is a coarse-grained rate multiplier built from a WKB suppression and an effective amplitude scale. The WKB action entering r_N is

$$S_N = \int_{\mathcal{B}_N} \sqrt{(V_{\text{eff}} - \omega_N^2)_+}\, \mathrm{d}s. \tag{22}$$

4.1 Two-center reduction and Feshbach interpretation

The rank-2 form in Eq. (19) can be read as a controlled ansatz for a two-lobe collective subspace. Let $P_N^{(2)}$ project onto the two localized lobe states $|N,+\rangle$ and $|N,-\rangle$, and let $Q_N^{(2)} = 1 - P_N^{(2)}$. If the eliminated complement has a stable gap, a Feshbach–Schur reduction [15] gives an effective operator

$$K_N^{\text{eff}}(E) = P_N^{(2)} K_N P_N^{(2)} - P_N^{(2)} K_N Q_N^{(2)} (Q_N^{(2)} K_N Q_N^{(2)} - E)^{-1} Q_N^{(2)} K_N P_N^{(2)}. \tag{23}$$

The release rank-2 matrix is the finite-dimensional closure of this structure after the diagonal lobe rates and off-diagonal coherent mixing are compressed to $\Gamma_{N,a}$, $\Gamma_{N,b}$, and χ. The short article does not require a full microscopic derivation of K_N, but it does require the rank-2 compression to be stated as an assumption rather than smuggled in as a theorem.

4.2 Effective phase-space and visibility weights

The release g_N is best read as an effective shell-volume weight. Legacy Cardy-like or holographic motivations remain useful background, but the baseline number entering Eq. (17) is the bounded two-dimensional phase-space profile audited in the long companion. Likewise, B_N is a visibility factor: it controls how strongly an occupied layer appears in the chosen visible-sector diagnostic. The neutral choice $B_{N>3} = 1$ is important because it prevents the absence of a fourth visible layer from being engineered by a visibility cutoff. In the release interpretation, high-N control comes from the phase-space/tail prescription and the independent finite-domain kinetic certificate, not from hiding extra layers in B_N.

Figure 3 gives the physical picture: low layers are localized in the double-well structure, while higher layers approach the continuum.

5 Main Release Results

The long technical companion records the full numerical audit trail. The short-paper result set should be restricted to the minimal release observables in Table 4. The numerical values below are the current gated release values generated by `code/generate_prd_short_figures.py` and recorded in `paper_prd/figures/prd_short_release_numbers.json`.

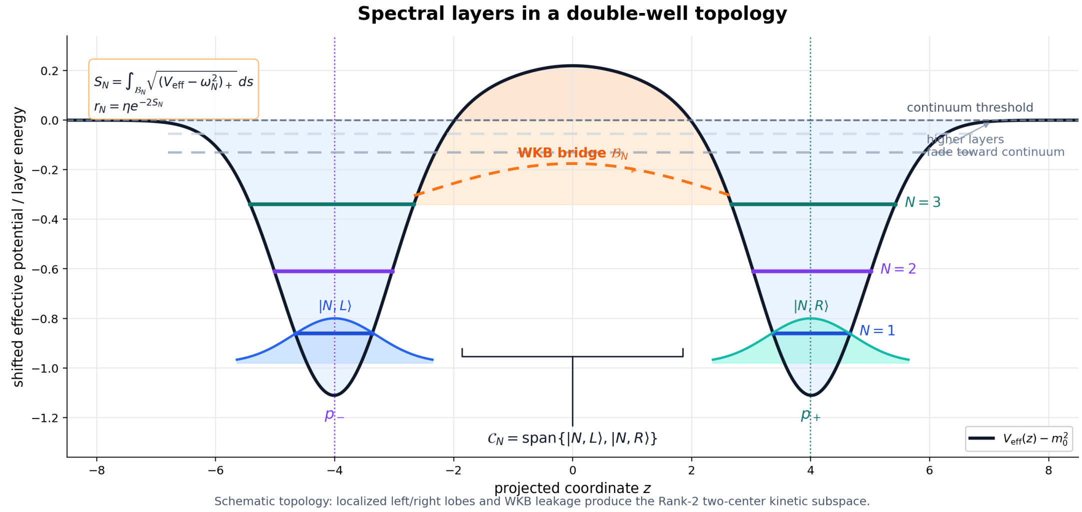

Figure 3: Spectral layers in a double-well topology. The rank-2 matrix in Eq. (19) is the effective two-lobe description of localized modes, with WKB suppression controlled by Eq. (22).

Table 4: Minimal release-result table for the short article. The numerical entries are generated from the canonical release artifacts by `code/generate_prd_short_figures.py`.

Observable	**Current release statement**	**Short-paper interpretation**
First-three occupancy	$\mathcal{R}_3 > 0.90$ on 92.7222% of the sampled release grid and $\mathcal{R}_3 > 0.95$ on 92.6389%.	Evidence for first-three-layer concentration within A1–A5, not a proof of exactly three SM generations.
No-fourth kinetic layer	On the 8 audited $D_{\rm sep}$ values, $n_{\rm bound} \in [1, 3]$ and max `supports_N4_bound` $= 0$.	A finite-domain threshold certificate, not a continuum theorem.
Table III WKB consistency	The canonical gate gives $\max_D \lvert \log r_1(D) + 2S_1(D)\rvert = 2.18202 \times 10^{-4}$ for displayed rounded values.	A numerical consistency gate for the WKB action/tunneling table.
Flavor kernel	The matched $H \to \mu\mu$ kernel is row-stochastic and non-idempotent, not a projector.	Wording and positivity are controlled by the release guard.
$H \to \mu\mu$ diagnostic	For $\mu_{\mu\mu}^{\rm obs} = 1.4 \pm 0.4$, $f(\chi^2_{\mu\mu} \leq 4) = 0.1500$, with best $\chi^2 = 2.20798 \times 10^{-5}$.	Target-region visualization only; no formal global likelihood is claimed.

5.1 Reading the occupancy map

The release map in Fig. 4 should be read in two layers. The winner panel records which N has the largest normalized occupancy P_N at each sampled (D_{sep}, η). The $\mathcal{R}_3$ panel records the cumulative first-three weight

$$\mathcal{R}_3(D_{\text{sep}}, \eta) = P_1(D_{\text{sep}}, \eta) + P_2(D_{\text{sep}}, \eta) + P_3(D_{\text{sep}}, \eta). \tag{24}$$

The headline statement is that $\mathcal{R}_3 > 0.90$ on 92.7222% of the release grid and $\mathcal{R}_3 > 0.95$ on 92.6389% of the release grid. This is an occupancy statement. It does not assert that exactly three bound eigenvalues exist at every D_{sep}, nor does it assert that a microscopic chiral family-count theorem has been proved.

5.2 WKB consistency gate

The WKB action table is a useful example of the release philosophy. The displayed S_1 and r_1 values are accepted only if the canonical artifact satisfies

$$\max_{D_{\text{sep}}} |\log r_1(D_{\text{sep}}) + 2S_1(D_{\text{sep}})| < \varepsilon_{\text{gate}} \tag{25}$$

after the same rounding conventions used in the paper. The current displayed residual is 2.18202×10^{-4}, so the table is treated as a numerical-consistency pass rather than a hand-copied result. This gate is intentionally mundane. It protects the more interesting physics discussion from avoidable arithmetic drift between the manuscript, the generated table, and the release artifact.

5.3 No-fourth kinetic certificate

The no-fourth result is similarly narrow. The audited single-track kinetic proxy covers 8 displayed D_{sep} values, with negative bound-mode count ranging from 1 to 3, and with maximum fourth-layer support indicator 0. The correct interpretation is

$$\text{audited finite-volume kinetic proxy} \Longrightarrow \text{no certified fourth bound layer on the tested domain.} \tag{26}$$

It is not

$$\text{full continuum EYMH theory} \Longrightarrow \text{exactly three SM generations.} \tag{27}$$

This wording preserves the useful threshold information without making the finite-volume certificate carry a claim it was not designed to prove.

5.4 Numerical gate summary

The release gates used by the short article are deliberately simple enough to be repeated during submission packaging:

Table 5: Minimal executable gates behind the short-article results.

Gate	**Purpose**	**Short-paper role**
WKB table gate	Checks Eq. (25).	Protects WKB table arithmetic.
Fig. 05 parity gate	Compares live Fig. 05 path to registered fractions.	Protects $\mathcal{R}_3$ headline values.
Short-article figure generator	Regenerates short-article figures and release macros.	Keeps table/figure numbers non-hand-entered.
Short-article LaTeX build	Compiles the submission draft.	Protects page count and layout.

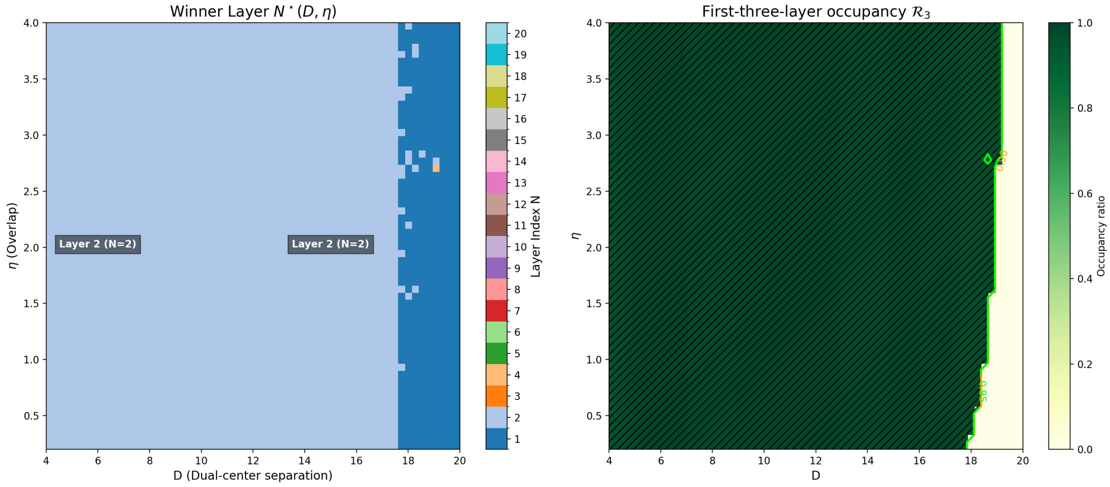

Figure 4: Release first-three-layer occupancy map. The left panel shows the winner layer on the release scan, while the right panel shows $\mathcal{R}_3 = P_1 + P_2 + P_3$. The canonical Fig. 05 parity gate requires the live plotting path to reproduce the registered fractions before the map is accepted.

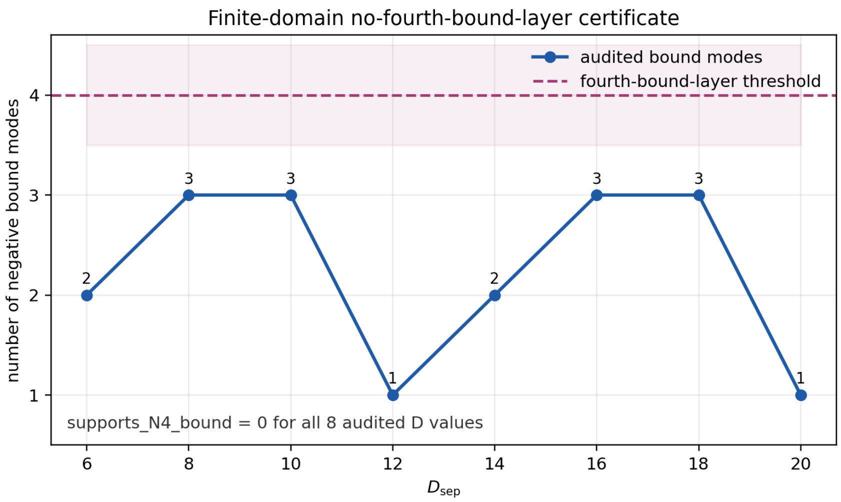

Figure 5: Finite-domain no-fourth-bound-layer certificate. The audited single-track kinetic proxy has one to three negative bound modes over the displayed D_{sep} values, but no certified fourth bound layer. This is a finite-domain statement, not a continuum theorem.

6 Projector Coherence Along D_{sep}

A useful optional strengthening is the spectral-projector transport theorem along D_{sep}. The point is modest but important: the low-mode labels $N = 1, 2, 3$ should be treated as transported projectors of a parameter-dependent operator family, not as unrelated pointwise eigenvectors.

Let $H(D_{\mathrm{sep}})$ be the finite-volume self-adjoint operator associated with Eq. (10), after the coordinate pullback to a fixed reference domain. If a Riesz contour $\Gamma_N(D_{\mathrm{sep}})$ separates the N-th eigenvalue from the rest of the spectrum, then

$$P_N(D_{\mathrm{sep}}) = \frac{1}{2\pi i}\oint_{\Gamma_N(D_{\mathrm{sep}})} (z - H(D_{\mathrm{sep}}))^{-1}\,\mathrm{d}z \tag{28}$$

varies continuously, and differentiably when $H'(D_{\mathrm{sep}})$ exists in operator norm [16]. Differentiating the resolvent identity gives the standard expression

$$P_N'(D_{\mathrm{sep}}) = \frac{1}{2\pi i}\oint_{\Gamma_N(D_{\mathrm{sep}})} (z - H)^{-1} H'(D_{\mathrm{sep}})(z - H)^{-1}\,\mathrm{d}z, \tag{29}$$

where $H = H(D_{\mathrm{sep}})$ inside the integrand. If the contour has length $|\Gamma_N|$ and distance at least δ_N from the rest of the spectrum, then the elementary bound

$$\|P_N'(D_{\mathrm{sep}})\| \leq \frac{|\Gamma_N|}{2\pi}\frac{\|H'(D_{\mathrm{sep}})\|}{\delta_N^2} \tag{30}$$

follows immediately. This bound is analytically clean but often numerically pessimistic because $\|H'\|$ is dominated by the regulated Plummer core rather than by the low-mode matrix elements.

The T-series audit therefore separates three levels. T2 gives an explicit analytic envelope for $\|H'(D_{\mathrm{sep}})\|$ from the derivative of the two-center Ω. T3a replaces the crude sup-norm route by weighted Hellmann–Feynman matrix elements in the low-mode sector. T3c performs the literal variable-box pullback $z = L(D_{\mathrm{sep}})y$, with $L(D_{\mathrm{sep}}) = D_{\mathrm{sep}}/2 + 6$, so that changes of the axial box, kinetic scaling, measure, and boundary terms are tested on the same reference coordinate. The current T1–T3c package supports this statement on the audited finite grid, with the strongest numerical evidence coming from the literal variable-box coordinate pullback. This package remains optional; it does not change the baseline map. In the current certificate, T3a gives $\max_n \|P_n'\| \leq 0.084095045889449382$, while T3c gives adjacent rank-one projector drift at most 0.013928181522331775, first-three cluster drift at most 0.011846382262997952, and pullback-versus-G1 low-three eigenvalue drift at most 8.02574×10^{-4}.

The conceptual payoff is limited but useful. It allows the labels $N = 1, 2, 3$ to be discussed as coherent low-mode projectors along the scanned D_{sep} interval, provided the certified gaps remain open. It does not convert the occupancy closure into a chiral family-count theorem, and it does not retune the release map.

7 Higgs-Diagnostic Map

The $H \to \mu\mu$ comparison is retained only as a reference-normalized scan diagnostic. Schematically,

$$\mu_{\mu\mu}^{\mathrm{pred}}(D_{\mathrm{sep}}, \eta) = \left|\frac{C_{eH}^{\mu\mu}(D_{\mathrm{sep}}, \eta)}{C_{eH}^{\mu\mu}(D_{\mathrm{sep}}^0, \eta^0)}\right|^2 \left[\frac{\Gamma_{\mathrm{tot}}(D_{\mathrm{sep}}, \eta)}{\Gamma_{\mathrm{tot}}(D_{\mathrm{sep}}^0, \eta^0)}\right]^{-1}. \tag{31}$$

The reference point $(D_{\mathrm{sep}}^0, \eta^0)$ is a normalization convention, not an independent derivation of the absolute muon Yukawa coupling. We therefore describe this comparison as a reference-normalized diagnostic target region rather than experimental validation.

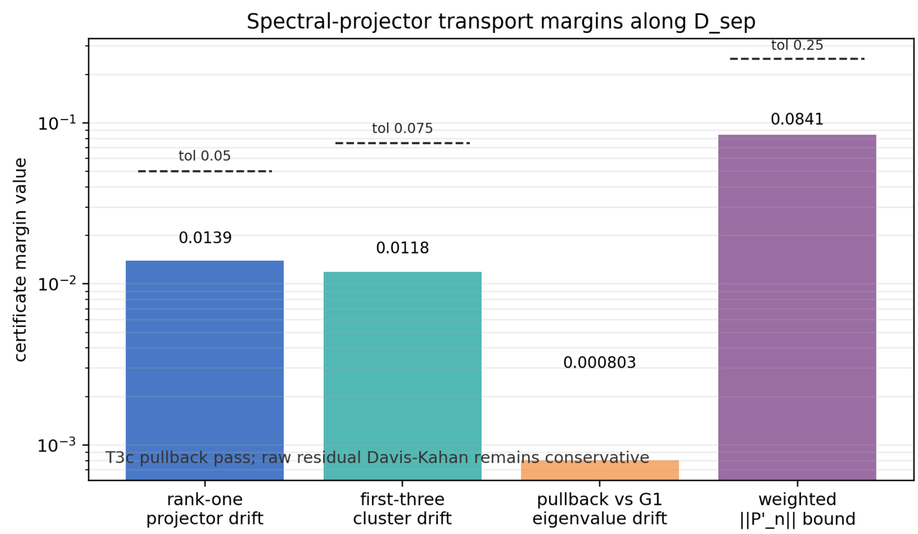

Figure 6: Optional spectral-projector transport margins along D_{sep}. The certified object is the actual variable-box coordinate-pullback projector step; the raw residual Davis–Kahan comparator remains intentionally conservative and is not used as the pass criterion.

The flavor-smearing object used in this diagnostic is a nearest-neighbor row-stochastic kernel,

$$K_{\text{fl}}(\epsilon_{\text{mix}}) = \begin{pmatrix} 1-\epsilon_{\text{mix}} & \epsilon_{\text{mix}} & 0 \\ \epsilon_{\text{mix}}/2 & 1-\epsilon_{\text{mix}} & \epsilon_{\text{mix}}/2 \\ 0 & \epsilon_{\text{mix}} & 1-\epsilon_{\text{mix}} \end{pmatrix}. \tag{32}$$

It is not a projector:

$$K_{\text{fl}}^2 \neq K_{\text{fl}} \tag{33}$$

for generic ϵ_{mix}. Its eigenvalues are

$$1, \qquad 1-\epsilon_{\text{mix}}, \qquad 1-2\epsilon_{\text{mix}}, \tag{34}$$

so the release positivity guard keeps $\epsilon_{\text{mix}} \leq 1/2$. This wording is deliberately more prosaic than "projector" because the matrix is a bounded mixing kernel, not an idempotent projection onto a physical subspace.

The observed ATLAS input is used only through the one-dimensional target value $\mu_{\mu\mu} = 1.4 \pm 0.4$ [13]. The short article therefore reports the fraction of scan points in a simple $\chi^2 \leq 4$ target band and the best grid value, but it does not claim a look-elsewhere-corrected exclusion, a global posterior, or an absolute Standard Model EFT Wilson coefficient measurement. For the fixed reference-normalization point used in the release figures, the target-band fraction is $f(\chi^2_{\mu\mu} \leq 4) = 0.1500$. The long companion's reference-normalization audit varies the reference point over 53 anchors and gives

$$f(\chi^2_{\mu\mu} \leq 4) \in [0.0333,\, 0.2411]. \tag{35}$$

This spread is not a failure of the diagnostic; it is the reason the short article treats Fig. 7 as a target-region visualization rather than as experimental support for a unique absolute normalization. Turning Fig. 7 into a genuine prediction would require an independent normalization theorem for $C^{\mu\mu}_{eH}$, a nuisance-parameter model, and a pre-specified likelihood.

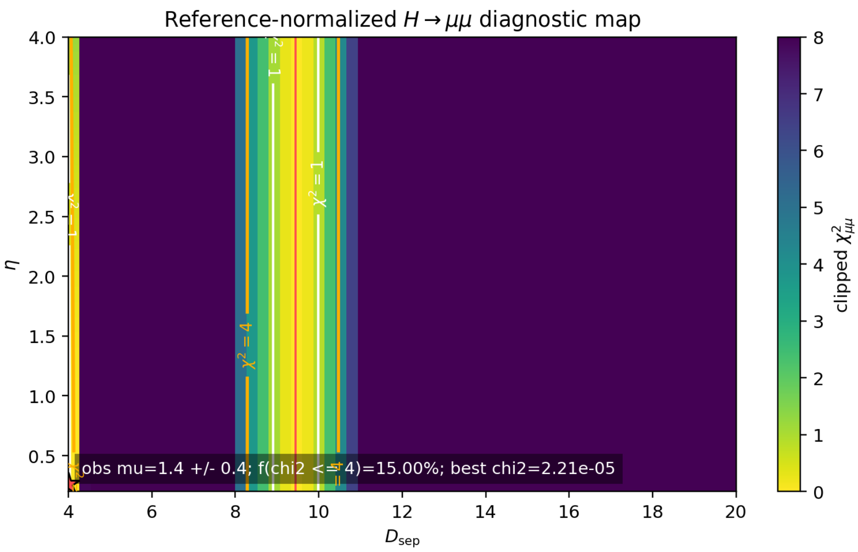

Figure 7: Reference-normalized $H \to \mu\mu$ diagnostic map. The color scale shows the clipped scan-level $\chi^2_{\mu\mu}$, the contours mark $\chi^2 = 1$ and $\chi^2 = 4$, and the star marks the best grid point. This figure visualizes a target region only; it is not a formal global likelihood or an absolute Higgs-sector prediction.

8 Limitations

The present closure has four front-facing limitations. First, the projected two-center background is fixed rather than derived from a complete EYMH parent. Second, the spectral label N is used as an EFT generation-layer label, while a protected chiral-family theorem remains a conditional companion route. Third, the no-fourth statement is finite-domain and operator-chain specific. Fourth, the Higgs comparison is reference-normalized and cannot be read as an absolute prediction without an independent normalization theorem.

These limitations are not cosmetic; they define the honest boundary of the current release. They are also what makes the short article safer for journal review: the core claim is computational and conditional, while stronger first-principles completions are separated into the supplement or future work.

8.1 What would close the limitations

The first limitation would be closed by deriving Eq. (9), including the positive regulated two-source density and C^2 control of Ω, from a parent variational or constraint problem. The C^2 requirement is not optional because $V_{\rm eff}$ depends on both Ω and $\nabla^2\Omega$.

The second limitation would be closed by a bridge theorem identifying the current transported low-mode projectors with protected chiral fermion zero modes. The optional Spin^c package gives a conditional template, but it requires geometric and no-exotics assumptions that are not consequences of the present scalar spectral closure.

The third limitation would be closed by replacing the finite-volume no-fourth certificate with a continuum-tail and $D_{\rm sep}$-continuous theorem for the same operator class. That is a strengthening of the current certificate, not a reason to reopen high-N visibility damping or introduce a new family.

The fourth limitation would be closed by an absolute matching theorem connecting the release $C^{\mu\mu}_{eH}$ kernel to a standard SMEFT normalization without using the same $H \to \mu\mu$ measurement as the reference point. Until such a theorem exists, the Higgs map remains a diagnostic.

9 Reproducibility and Supplement Map

The long manuscript and repository artifacts serve as the technical companion for this short article. Before using the companion, the reader should distinguish the status tags in Table 6. The intended topical split is then shown in Table 7.

Table 6: Status legend for the long technical companion.

Status tag	**Meaning**	**Effect on the short-article claim**
Baseline input	Used directly in the release numbers reported in the short article.	May support headline statements when A1–A5 and the stated audit domain apply.
Certificate	Proves or numerically certifies a finite-domain statement used to guard the baseline.	Supports robustness, but does not automatically globalize the claim.
Diagnostic / monitor	Checked for sensitivity, mechanism insight, or reviewer transparency but not propagated into the release baseline.	Cannot be used as a new input without a separate adoption gate.
Negative result	A route that failed its adoption gate or was closed as too weak.	Preserved to prevent repeated searches and selective reporting.
Future route	A possible strengthening, such as a parent derivation or protected-index completion.	Not assumed by the present release.

Table 7: What remains in the short-article paper versus the long technical supplement.

Topic	**Short article**	**Supplement / technical companion**
Projected operator	Definitions, conformal reduction, two-center figure.	Full derivation checks, table generation, numerical WKB audit.
Occupancy closure	Master equation, rank-2 matrix, release result table.	Full proposition chains, branch histories, negative families.
Open-system branch	One diagnostic-status sentence.	O-series exact-bridge certificates and reservoir audits.
Static-width branch	Local-theorem status only.	CAP/ECS/Whittaker/DtN/S-series certificates.
Protected Spin^c route	Future/conditional route only.	V0–V6 conditional bridge package.
DSS/Floquet route	Closed diagnostic/negative note only if mentioned.	D0–D3 feasibility and failure artifacts.
Projector transport	Optional theorem sketch and final numerical margins.	T1–T3c scripts, CSVs, and detailed proof note.

A final submission package should include exact commit hashes, artifact checksums, and one-command reproduction instructions. The short article should cite the supplement by version rather than embedding long internal path names in prose.

9.1 Minimal reproduction commands

The short-article build is intentionally much smaller than the full technical companion. The minimal command sequence is:

```
cd paper_epjc
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cd ..
python3 code/check_table_iii_wkb_consistency.py
python3 code/check_fig05_r3_parity.py
```

The first command builds the EPJC-preparation PDF from the copied release figures, JSON, macros, and release table. The last two commands verify the canonical WKB relation $r_1 = \exp(-2S_1)$ and the live plot-generator parity for the $\mathcal{R}_3$ map.

For the final submission archive, the generated PDF should be paired with the source commit, the release-number JSON, and checksums for the figures and generated TeX fragments. This is preferable to embedding long artifact paths inside the main text. The reader sees the physical statement in the article; the supplement and registry preserve the audit trail.

9.2 Data, code, and supplement availability

For an EPJC-style submission, the article should be accompanied by a versioned public repository archive containing the source commit, the generated release-number JSON, the figure-generation inputs, and the finite-domain certificate artifacts used in Tables 4 and 5. The long technical manuscript should be cited as a supplementary audit companion, not as a second independent baseline-claim paper. At final submission time, the repository archive should preferably be assigned a persistent identifier, and the manuscript should state that all numerical claims in the short article can be regenerated from the archived scripts and artifacts. This availability statement is part of the claim calibration: reproducibility supports the conditional closure, but it does not turn assumptions A1–A5 into first-principles theorems.

10 Conclusion

The short-article version of PSLT should be a calibrated computational-physics manuscript: it defines a projected two-center spectral problem, a rank-2 occupancy closure, and a release-grid diagnostic showing first-three-layer concentration under stated assumptions. The long paper remains valuable, but its role is technical certification rather than front-door exposition. This split gives reviewers a clear target: evaluate the conditional EFT closure and its reproducibility, without being asked to accept every optional mechanism branch as part of the main claim.

The most important scientific output of the short version is therefore not a slogan about "explaining three generations." It is a reproducible map from a projected two-center operator to a normalized spectral-layer occupancy distribution, together with explicit gates showing where the current closure is robust and where it is only diagnostic. If future work promotes A1 or A2 to parent-side theorems, the same short-article architecture can absorb that strengthening. Until then, the defensible journal claim is the conditional one stated in Eq. (3).

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